COLLEGE OF ENGINEERING KOTTARAKKARA

MANAGED BY IHRD, GOVT. OF KERALA

ETC P.O, Kottarakkara, Kollam, Kerala – 691531

CONFERENCE PROCEEDINGS

National Conference on Multidisciplinary Approaches in AI & Sustainable Engineering

NCMASE'26
CONFERENCE PROCEEDINGS

National Conference on Multidisciplinary Approaches in AI & Sustainable Engineering

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INSTITUTION BACKGROUND

The College of Engineering Kottarakkara (CEK) is a premier technical institution established in 2004 under the Institute of Human Resources Development (IHRD), Government of Kerala. The college is approved by the All India Council for Technical Education (AICTE), New Delhi, and is affiliated to APJ Abdul Kalam Technological University (KTU).

Located in Kottarakkara, Kollam, CEK is well-connected in terms of transportation, communication, and accommodation. The campus is part of the National Knowledge Network (NKN) and features a digitally enabled learning environment with smart classrooms and state-of-the-art ICT facilities. Admissions are based on ranks secured in the All-Kerala Common Entrance Examinations.

The college promotes a vibrant academic culture through student clubs, technical forums, innovation cells, and professional chapters including IEEE, ISTE, and CSI. The Innovation and Entrepreneurship Development Cell (IEDC) fosters creativity, leadership, and entrepreneurial skills. The college continually strives for excellence in teaching, research, and development.

VISION

To emerge as a center of excellence in engineering education and research, fostering innovation, ethical values, and technological competence to produce globally responsible professionals capable of contributing to the sustainable development of society.

MISSION

1. To provide high-quality technical education that nurtures creativity, critical thinking, and professional competence.
2. To develop ethical, skilled engineers capable of addressing contemporary global challenges.
3. To promote research, innovation, and industry collaboration through advanced learning environments.
4. To foster holistic student development, leadership, and social responsibility.

INTRACRANIAL ANEURYSM DETECTION AND SEGMENTATION USING DEEP LEARNING - A GLANCE INTO THE METHODS, METRICS AND CHALLENGES

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ABSTRACT

Intracranial aneurysms (IAs) are pathological dilations of cerebral arteries that pose a significant health risk, affecting 3–6% of the adult population. Rupture leads to subarachnoid hemorrhage with mortality rates exceeding 40%. Early detection through non-invasive imaging is crucial for timely intervention. This comprehensive survey reviews recent advances in deep learning-based approaches for automated IA detection and segmentation from medical imaging modalities including Computed Tomography Angiography (CTA), Magnetic Resonance Angiography (MRA), and Digital Subtraction Angiography (DSA). We systematically analyze methodologies across three categories: detection-focused methods, segmentation approaches, and hybrid detection-segmentation systems. Key architectures reviewed include 3D Convolutional Neural Networks (CNNs), U-Net variants, global-local networks, and ensemble methods. Through comparative analysis of over 10 state-of-the-art models including GLIA-Net, HeadXNet, and specialized MRA detection systems, we examine their architectural innovations, training strategies for handling class imbalance, dataset requirements, and performance metrics (sensitivity: 82–95%, false positives: 0.1–4.4 per case). Current challenges including small aneurysm detection, multi-center generalization, and clinical workflow integration are highlighted along with future research directions toward explainable AI and real-time processing capabilities.